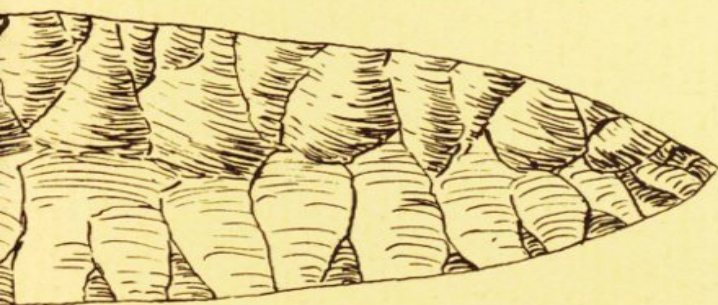
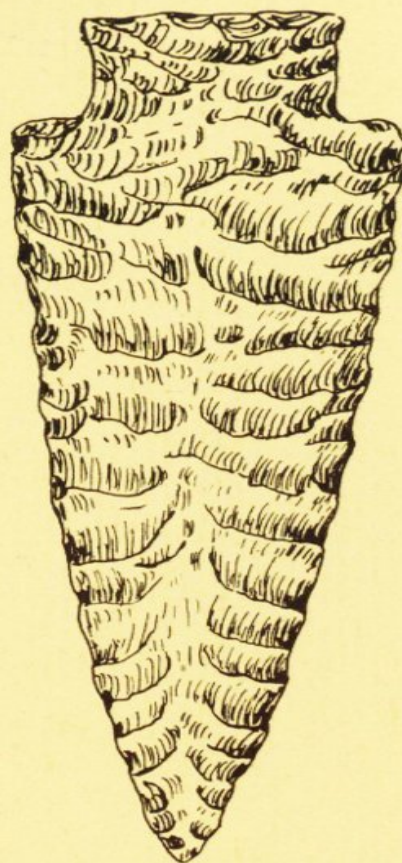
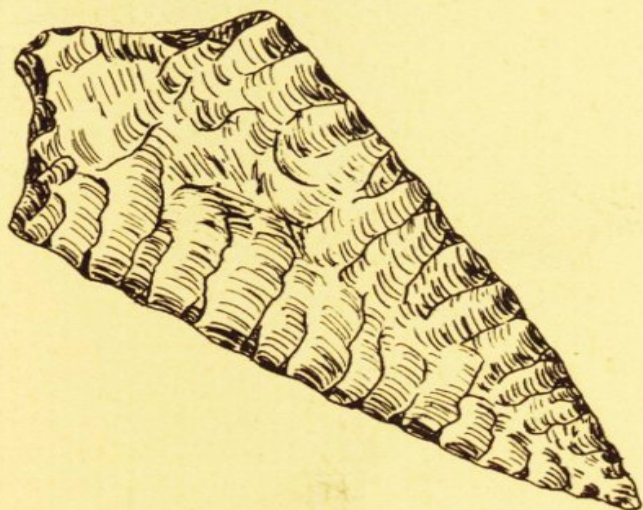
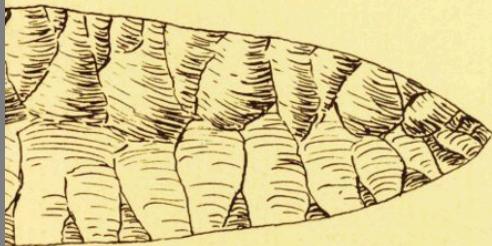
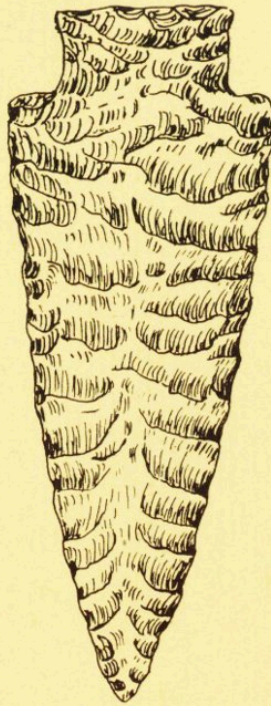
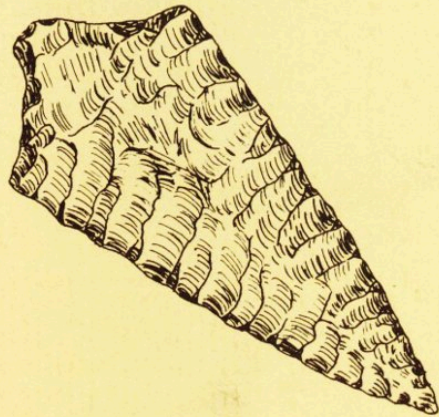




Early Man
Projectile Points
in the Southwest

MUSEUM OF NEW MEXICO PRESS
POPULAR SERIES PAMPHLET NO. 4



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*** START OF THE PROJECT GUTENBERG EBOOK EARLY MAN
PROJECTILE POINTS IN THE SOUTHWEST ***

Early Man Projectile Points in the Southwest

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EARLY MAN PROJECTILE POINTS IN THE SOUTHWEST
by Kenneth Honea

INTRODUCTION

The cultures of Early Man in the Southwest, though yet imperfectly known, seem most readily distinguishable by characteristic types of projectile points. The majority of finds, and they are rare, have been made on the surface; much less often at camp or kill sites.

Camp sites were situated on ridges, in sand dunes, or on hills overlooking streams, lakes, or ponds. Occupational features generally include hearths, split and charred food bones, debris from stone tool manufacturing, and a full inventory of stone tools. Seeds, grinding stones, storage and cooking pits are rarely found at such early sites.

Kill sites, by contrast, are usually situated on the edges of streams, lakes, or ponds. Features will include animal skeletons, sometimes partially dismembered, points used in killing game, some stone butchering tools such as scrapers, knives, utilized flakes, and waste flakes resulting from the sharpening of stone tools. Preferred parts of game may be missing, indicating they were carried back to camp. Hearths are occasionally found at kill sites, indicating that part of the game may have been prepared there.

Points used by Early Man were likely thrown at game with a spear-thrower or atl-atl, rather than shot at them with a bow and arrow. Atl-atls were probably of wood, and had a long groove on one side. Into this groove was placed the spear with a point hafted onto one end. The atl-atl was thrust back and thrown forward, releasing the spear on the forward thrust.

On the basis of findings, archaeologists have reconstructed two widespread methods of hunting by Early Man. In the first instance, game was observed, stalked and killed at watering places. In the second, game was driven into an arroyo, stream, lake, or pond, or stampeded over a cliff.

EARLY MATERIALS AND TECHNOLOGIES

It is a curious fact that Early Southwestern Man most often made his tools of very finely textured stone. Perhaps this custom was induced by the exacting, finely controlled technologies of stone flaking practiced and the kinds of tools made. Indeed, workmanship on many projectile points, knives, and scrapers is so well achieved that one is led to believe that Early Man strove to express some degree of esthetic idealism in his tools.

Demands in technology often led to the widespread trading of choice materials. The best known of these is from the alibates quarries near Amarillo in the Texas Panhandle. Fine quality flint from this locality has been found at Early Man sites as far west as Arizona and as far north as Wyoming and Montana.

Early man prepared point “blanks” by the striking of suitable primary flakes by direct percussion from a block of stone, the core. The striker consisted either of a hard subround hammerstone, or a cylinder-hammer of soft stone, bone, antler, or hard wood. Flakes driven off by the former were usually thicker at one end. Those produced in the second instance were relatively thin throughout their length. Long flakes or “blades” were also often used as blanks, especially in the making of long projectile points. They could be detached from cores with either of the above strikers, or by holding a bone or stone “punch” on an edge of a core and striking the top of the punch with a hammerstone. This is a form of indirect percussion. Blades were also produced by what has been called impulsive pressure. A bone, antler, or stone-tipped crutch was applied to an edge of a core, and pressed downwards with the chest, driving off a blade.

Often cores were pre-shaped before the striking or pressing off of flakes or blades by what is termed the “Levallois technique.” A block of stone was initially trimmed on both faces into a round, oval, or triangular shape. Either one or a series of flakes or blades were then driven off one or both faces of the core, using any of the above percussion or pressure instruments. A distinct advantage to the technique was that the upper face of the

detached flake or blade blanks had already been trimmed in the process of initial preparation of the core. This same preparation also predetermined to a large extent the shape of blanks produced in the Levallois technique.

The processes by which appropriate blanks were turned into finished 3 projectile points involved several manufacturing stages and one or a combination of flaking techniques. The techniques employed consisted of direct or indirect percussion, using a hammerstone, cylinder-hammer, or punch, and direct or indirect pressure, using a crutch, an animal claw or tooth, the edge of a length of bone, antler, or stone.

The principal manufacturing stages include: 1) over-all initial shaping by the careful thinning of both faces of the blank; 2) secondary trimming of lateral and basal edges, including eventual fluting; and 3) smoothing by grinding of lateral and basal edges.

As will later be seen in the descriptions, special kinds of initial flaking, or facial thinning, are often characteristically associated with certain Early Man points. On some, flaking may conform to traditional patterns, while on others it may be irregular. The following kinds of initial flake scars are distinguished:

Irregular: flake scars are of irregular shape, and they are in no particular alignment to one another or the point long-axis.

Transverse: long, parallel flake scars are horizontal, or nearly so, to the point long-axis; opposite flake scars end smoothly near the point center line, appearing to form a continuous flake scar across the point face.

Collateral: similar to transverse flaking, except that opposite flake scars end abruptly near the point center line, forming in some cases a central ridge.

Oblique: long, parallel flake scars are oblique or slanted in relation to the point long-axis.

BLANCO POINTS

To be counted possibly among the earliest projectile point types of the Southwest are what have been called Blanco points, named after the Blanco River of Central Texas. They were discovered by the writer, along with other stone and bone tools, in alluvial banks and other formations which in part may have accumulated during the Late Pleistocene. Lanceolate-shaped points from type localities are characteristically desert varnished and have been dulled and battered by being washed about in stream gravels. Heavily weathered points quite like those of the Blanco type have been reported in other parts of Texas and also in central and southeast New Mexico. They have also been recovered from a number of sites in Mexico and South 4 America under circumstances that seem to suggest great antiquity. The eminent American archaeologist, Marie Wormington, has suggested that points much like those of our Blanco type may well have been the basic type from which later Early Man points evolved.



Blanco

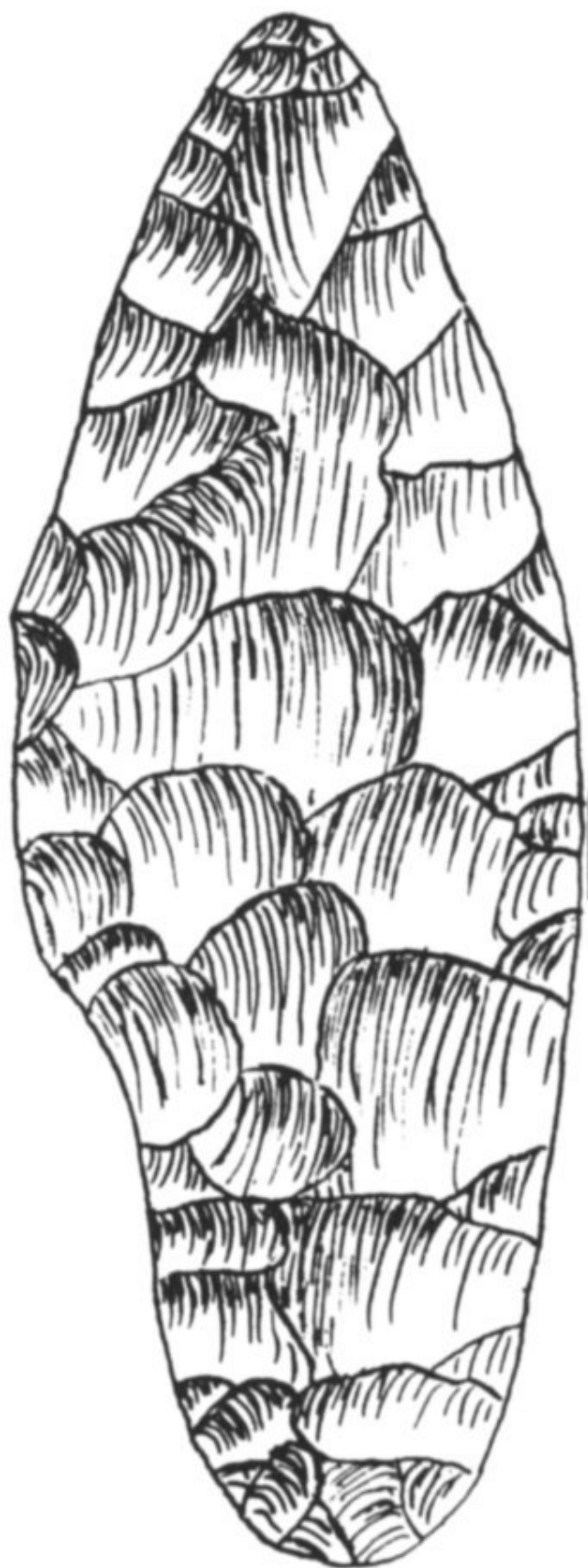
Blanco points are lanceolate-shaped and are broadest at or somewhat below mid-section. The cross-section at this point is a flattened oval shape. Lateral

edges are slightly rounded, converging to a point at the tip, while the base is round. The few known specimens range in length from about 4.0 to 6.0 cm., in width from 2.0 to 2.5 cm., and in thickness 0.7 to 1.0 cm. Specimens having dimensions somewhat in excess of those listed may be expected. Manufacture is by direct percussion and is not particularly well achieved nor patterned. Initial flake scars on point faces are irregular; point bases have been thinned by multi-directional flaking. Pressure working of edges is absent. Due to intense weatherings, it is not known whether Blanco points were basally ground.

Blanco points are associated in central Texas with the San Marcos Complex. They may represent a primary Early Man projectile point type. Undisturbed San Marcos sites have not been found as yet. Tools ascribed to this complex have been found only in secondary gravel beds at type localities along the Blanco River. It is conceivable that some of these formations may have accumulated during the Late Pleistocene. In view of all factors, dating of Blanco points relative to other Early Man points must await future excavations.

SANDIA I POINTS

This projectile point type was named after the type site, Sandia Cave, situated at the north end of the Sandia Mountains near Albuquerque, New Mexico. Such points have also been found at the Lucy Site in Central New Mexico, in Central Texas, Oklahoma, Missouri, Iowa, and possibly 5 Alabama, Alberta, and Saskatchewan.



Sandia I

Sandia I points are lanceolate in shape and may be somewhat asymmetrical in outline. They are distinctive in having a single shoulder on one side of the stem. Gently rounded lateral edges on the upper portion taper to a point, while the rounded stem edges taper to the somewhat narrow, rounded base. Specimens are broadest somewhat below mid-section and have a lenticular cross section at this point. Both the stem and the base have been smoothed by grinding. Total length may vary from about 6.0 to 9.0 cm., and width from 2.0 to 4.0 cm. Manufacture was evidently by direct percussion and not as refined as on some later Early Man Points. Initial flake scars on point faces are irregular; point bases have been thinned by multi-directional flaking. Pressure retouch is absent.

Sandia I projectile points are accepted by most archaeologists as among the oldest documented, though not reliably dated, point types of the Southwest and North America in general. Similarly, the relationship of Sandia points to Clovis and other Early Man point types has not yet been clearly established. That Sandia points are older than Folsom points does remain certain, however. They may also be older than Clovis points.

SANDIA II POINTS

Sandia II points were discovered together with Sandia I points at Sandia Cave near Albuquerque, New Mexico. They differ from Sandia I points in a number of respects.

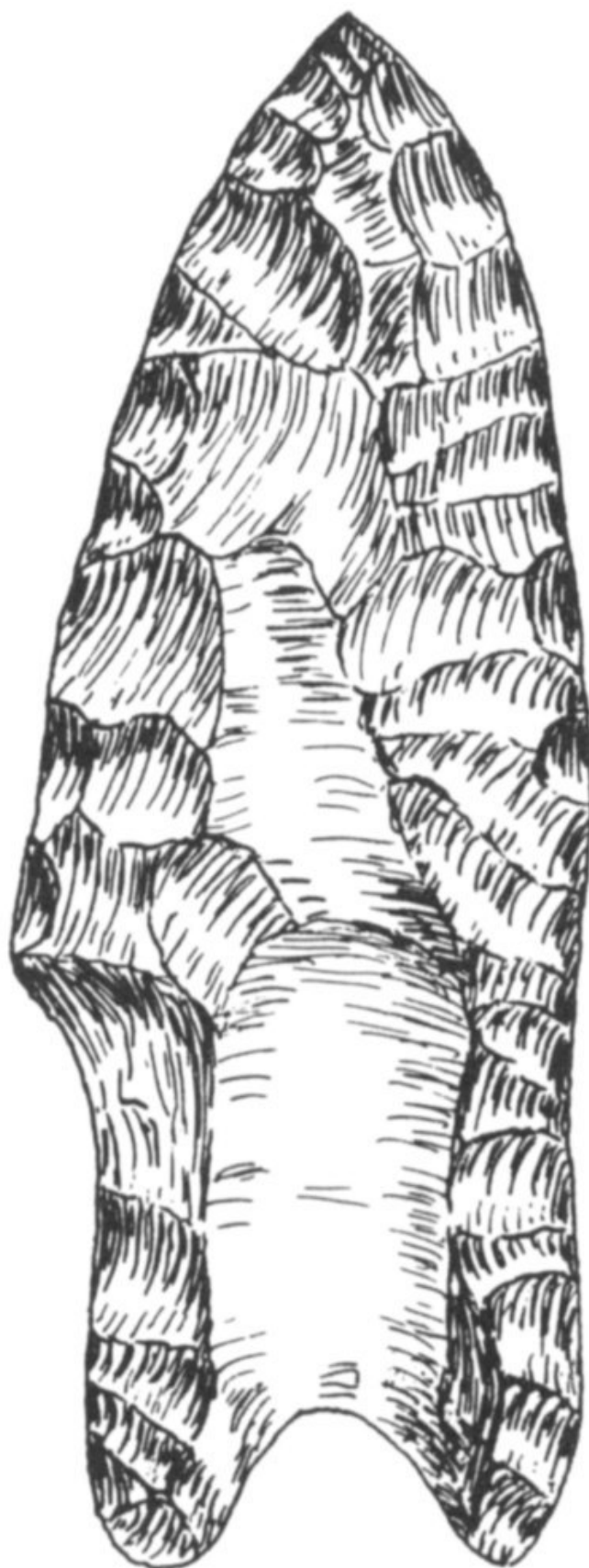
Single-shouldered Sandia II points are quite similar to those of the type Sandia I excepting details on the stem, base shape and certain aspects of workmanship. Instead of tapering to a rounded base, the straight-edged stem of Sandia II points contracts to a straight or slightly concave base. Workmanship is generally finer than in the other type, though achieved 6 by the same means. Resolved secondary edge thinning is noticeable on many specimens. Sandia II points are also marked by vertical basal thinning, produced by the removal of a large, or several smaller longish flakes parallel to the point long-axis, anticipating perhaps the fluting found on Clovis and Folsom points. Vertical basal thinning is absent on points of the Sandia I type.



Sandia II

SANDIA III POINTS

Sandia III points were discovered in 1954 together with Sandia I, Sandia II, and other Early Man points in a series of blow-outs near Lucy, central New Mexico.



Sandia III

Single-shouldered Sandia III points are similar in all respects to Sandia II points with these exceptions: the base on specimens of the Sandia III type is consistently deeply concave, and the stem portion is always vertically fluted.

CLOVIS POINTS

Clovis points are named after the type site located near Clovis, New Mexico. They were found associated with the remains of mammoth and other extinct animal forms in gravels at the bottom of a Late Pleistocene pond. This site is counted among the most important archaeological discoveries of the New World. The series of culture layers excavated from above the one containing Clovis material give the most complete sequence of Early Man cultures yet found at a single site.



Clovis

Clovis points are symmetrically lanceolate in outline and are broadest near mid-section or sometimes nearer to the tip. Upper lateral edges of the point body are slightly rounded. Lower lateral edges are commonly straight or slightly rounded and they may taper somewhat towards the consistently concave base. On some examples, the lateral edges near the base may be gently recurved. Both the basal concavity and the lateral edges are smoothed by intentional grinding. The characteristic flutes were struck vertically from only one or from both faces. In the latter case, they may not be of equal length. The longest flute usually does not extend up further from the base than about half the length of the point. Nonetheless, specimens are known on which fluting extends up almost the entire length of a face. Flaking on Clovis points was probably by direct percussion; lateral edges of some may have been thinned by secondary pressure retouch. Initial flake scars on point faces are irregular. Fluting was achieved by the removal of either one or a series of long flakes parallel to the point long-axis, possibly by indirect percussion. Dimensions are variable. Clovis points in the broader Southwest may range from nearly 3.5 to 14.0 cm. in length, 8 and in width from about 2.0 to 4.0 cm.

Radio carbon dates and geologic evidence obtained in recent years indicate Clovis points were in use by Early Man between about 11,000 and 11,500 years ago in the Southwest. Though older than Folsom points, there evidently was a period towards the end of the Clovis sequence, during which both Clovis and Folsom points were made and used. Clovis points have been found with the remains of extinct Pleistocene fauna, primarily mammoth, not only at the type locality, but at sites near Angus, Nebraska; Dent, Colorado; Miami, Texas; and Naco, Arizona. Finds of such points, mostly on the surface, have also been made in northern Mexico, California, on the eastern seaboard, in the Midwest, Canada, and Alaska.

Small Clovis points may be confused with Folsom points. However, Clovis points are both thicker and heavier; flutes do not ordinarily extend up the entire face, and workmanship is not as fine as on Folsom points. Also,

Folsom points often have a small nipple in the middle of the basal concavity, a feature lacking on Clovis points.

FOLSOM POINTS

Folsom points were first discovered in the twenties at an Early Man kill site near Folsom, in northeast New Mexico. The points were associated with the remains of an extinct form of giant Late Pleistocene bison. These finds were among the first to demonstrate the great antiquity of man in the New World.



Folsom

Folsom points are symmetrically lanceolate in shape, and are broadest at mid-section or somewhat nearer to the tip. They are fluted on either one or both faces; flutes usually extend up almost an entire face. Upper lateral edges are ordinarily gently rounded, while the lower lateral edges are usually straight and either parallel-sided or slightly contracting towards the

base. The base is consistently concave, in some examples, markedly so. Similar to Clovis points, stem edges near the base may be gently recurved. Average dimensions are: length 3.0 to 8.0 cm., but averaging 3.0 to about 5.0 cm., and width from about 1.5 to nearly 3.0 cm. Workmanship is 9 usually quite delicate, with initial shaping by either direct or indirect percussion or pressure. Initial flake scars on point faces, where preserved are irregular. Delicate secondary edge thinning was by pressure, during which the basal concavity was probably produced and a nipple left preparatory to fluting. Vertical fluting was probably produced by indirect percussion.

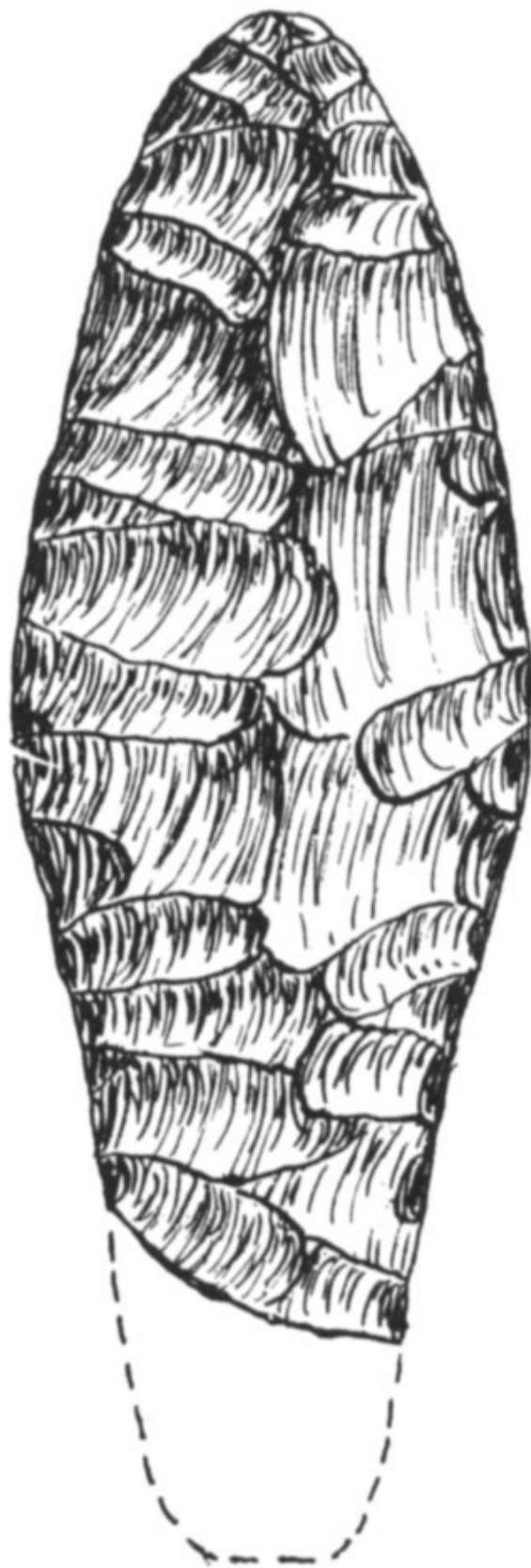
The latest research suggests that Folsom points may be dated between 10,000 and 11,000 years ago. Excavated sites producing Folsom points have in most all cases also produced the remains of giant bison, but not mammoth, which may have become extinct by the advent of Folsom Man. Folsom points have been discovered in both excavated and surface sites in most parts of the Southwest, including the Clovis type site in eastern New Mexico, numerous sites in most other portions of New Mexico, at the Lubbock, Lipscomb, Scharbauer, and Kincaid Sites in Texas, the Lindenmeier, Linger, and Powars Sites in Colorado, the Hell Gap Site in Wyoming and the MacHaffie Site in Montana.

HELL GAP POINTS

This recently defined Early Man projectile point has been named after a group of quarry sites in the Hell Cap Valley near Guernsey in east-central Wyoming.

Hell Gap points, similar to many other Early Man types, are essentially lanceolate-shaped. Upper lateral edges are slightly rounded, while lower lateral edges taper in straight lines to the generally narrow, slightly to noticeably rounded, occasionally nearly straight base. These points are commonly broadest at mid-section, but examples are known which are broadest a little below this point. Lower lateral edges are offset from the broadest section to such a degree that one is inclined to consider Hell Gap points stemmed. Dimensions are: length 4.5 to 8.8 cm., and width 2.0 to 3.0 cm. Workmanship is not nearly as fine as on Folsom points. Flaking was produced by direct percussion, with occasional secondary edge trimming by pressure. Initial flake scars on point faces are irregular; basal thinning was accomplished by multi-directional flaking. The stem and base are smoothed through grinding; basal smoothing may be absent on some specimens.

Surface specimens have been found in many parts of Wyoming, Montana, Colorado, New Mexico, and Arizona. Similar points are also known from California and parts of Texas. The most northerly known occurrence is Alberta, in Canada.



Hell Gap

MIDLAND POINTS

Midland points were first discovered at the Scharbauer Site near Midland, in West Texas. They were evidently contemporaneous with Folsom points, which they greatly resemble.



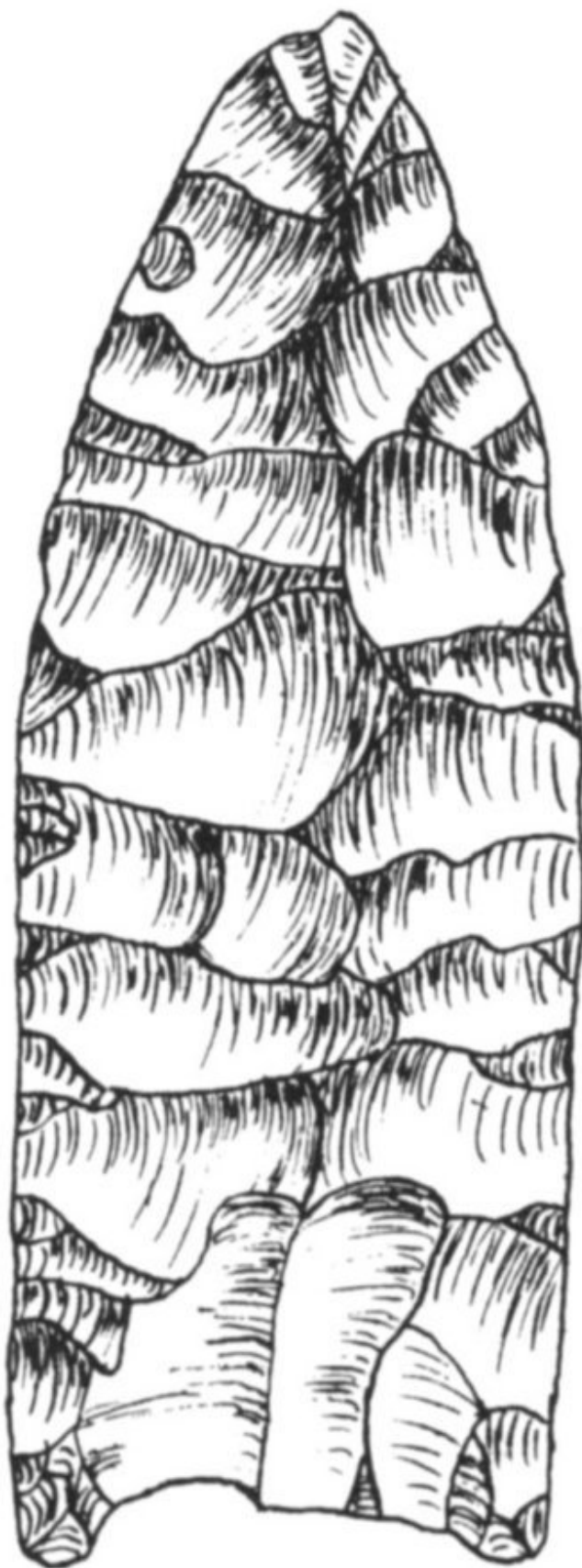
Midland

Formal description of Midland points has not yet been made. They are mentioned, however, as appearing a great deal like Folsom points in shape, size, and thinness, but are distinct from them in not being fluted. Otherwise they are apparently of the same fine workmanship and technological tradition as Folsom points.

Deposits in which a Midland point fragment was found with human remains at the Scharbauer Site have been dated to about 11,000 years ago, indicating a time range for this point type corresponding to that of Folsom. Similar points have been found near Lubbock, Texas; central New Mexico; and as far east as San Augustine County, Texas. Possibly they will be found to occur in other districts on or near the Llano Estacado. Quite frequently they are found together with Folsom points in west Texas and New Mexico localities. 11

PLAINVIEW POINTS

Plainview points were first defined on the basis of type specimens recovered from a bison kill site near Plainview in the Texas Panhandle.



Plainview

Lanceolate shaped, the upper lateral edges of the most Plainview points are slightly rounded but on some they are straight. Lower lateral edges are ordinarily straight and parallel-sided up to about one-half or three-quarters the distance from the base to the tip. They are usually broadest at mid-section. A variation in shape may include a recurve, rarely marked, of lateral edges near the base; such specimens are broadest either at or somewhat above the mid-section (see also the description of [Golondrina points](#)). The base of Plainview points varies from slightly to markedly concave; the former is typical. Workmanship is quite fine. Direct or indirect percussion or possibly pressure action is employed in retouching. Initial flake scars may be at either right angles or slightly oblique to the point long-axis. Though individual initial flake scars may be parallel, they are more often irregular. Dimensions of Plainview points are: length about 4.5 to 8.0 cm., width 1.8 to 2.8 cm. Bases of these points were sometimes thinned by the removal of the series of relatively large, longish flakes parallel to the point long-axis in a technique reminiscent of fluting. Others however, were basally thinned by the removal of flakes from the lower lateral edges at right angles to the point long-axis. Basal edges are usually smoothed.

Plainview points are considered by most archaeologists to date around 9,000 to 10,000 years ago. At both the type site and at the Lone Wolf Creek Site in Texas, they were associated with the remains of extinct bison.

MILNESAND POINTS

Milnesand points were first recorded at a site near Milnesand, New Mexico.



Milnesand

Lanceolate in shape, Milnesand points are broadest either at mid-section or somewhat above. Upper lateral edges are slightly to markedly rounded, while the lower edges are straight and taper a little towards the typically straight base. Bases on some may be slightly rounded or concave. Average dimensions are: length 5.0 to 8.0 cm., width 2.0 to nearly 3.0 cm. Workmanship is generally quite fine, and has been achieved with either direct or indirect percussion, or possibly pressure. Secondary thinning of both lateral and basal edges often occurs and is by pressure. Initial flaking of point faces is patterned: transverse flake scars are approximately parallel to one another and at right angles to the point long-axis. These scars terminate at the point mid-line; some specimens may have a distinct median ridge on one or both faces. Horizontal basal thinning, accomplished primarily during initial flaking may be accompanied by the removal of small flakes vertical to the point long-axis. Both the lower lateral edges and base are carefully smoothed.

Milnesand points at the type site were found with the remains of bison, possibly an extinct form. They occur widely in eastern New Mexico and western Texas. Others have been found in Nebraska, Iowa, Alberta, Saskatchewan, and Alaska. Similarities of Milnesand to Plainview points have been noted by archaeologists, who have suggested the two may be related and probably were partly contemporaneous. They may be distinguished from one another by the following attributes: The base on most Milnesand points is straight, while it is predominantly concave on Plainviews; thinning of Plainview bases was accomplished primarily by the removing of a few, fairly large vertical flakes, while Milnesand bases were thinned primarily by the removal of horizontal flakes, and only secondarily by the removal of small flakes. Finally, the smoothing of lower lateral edges on Milnesand points generally extends up further than on Plainview points, sometimes to a point beyond the mid-section.

MESERVE POINTS

Meserve points were first described from the Meserve Site near Grand Island, Nebraska. They are found in most parts of the Southwest.

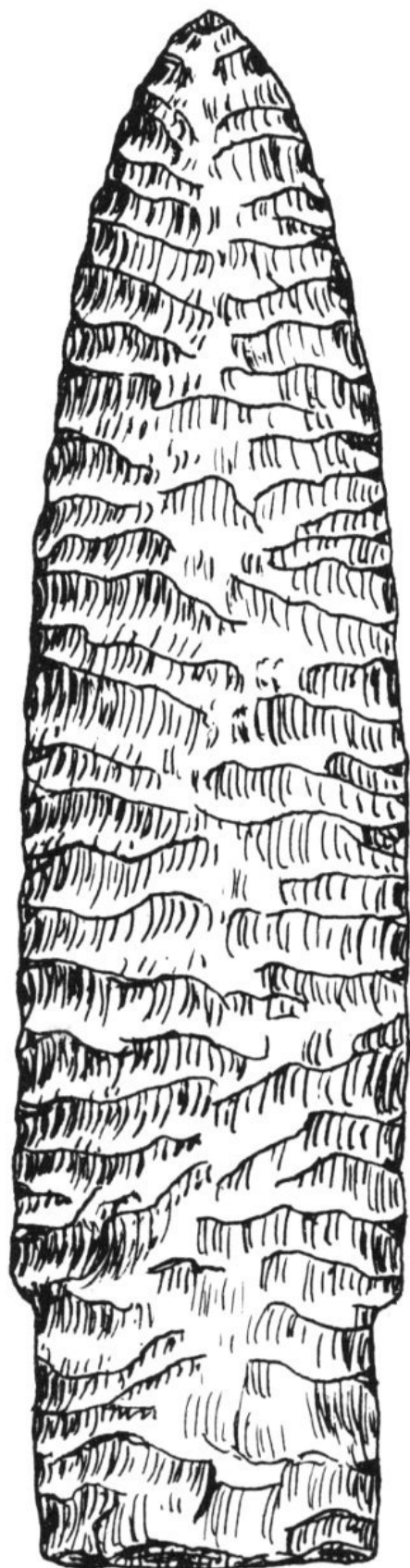


Meserve

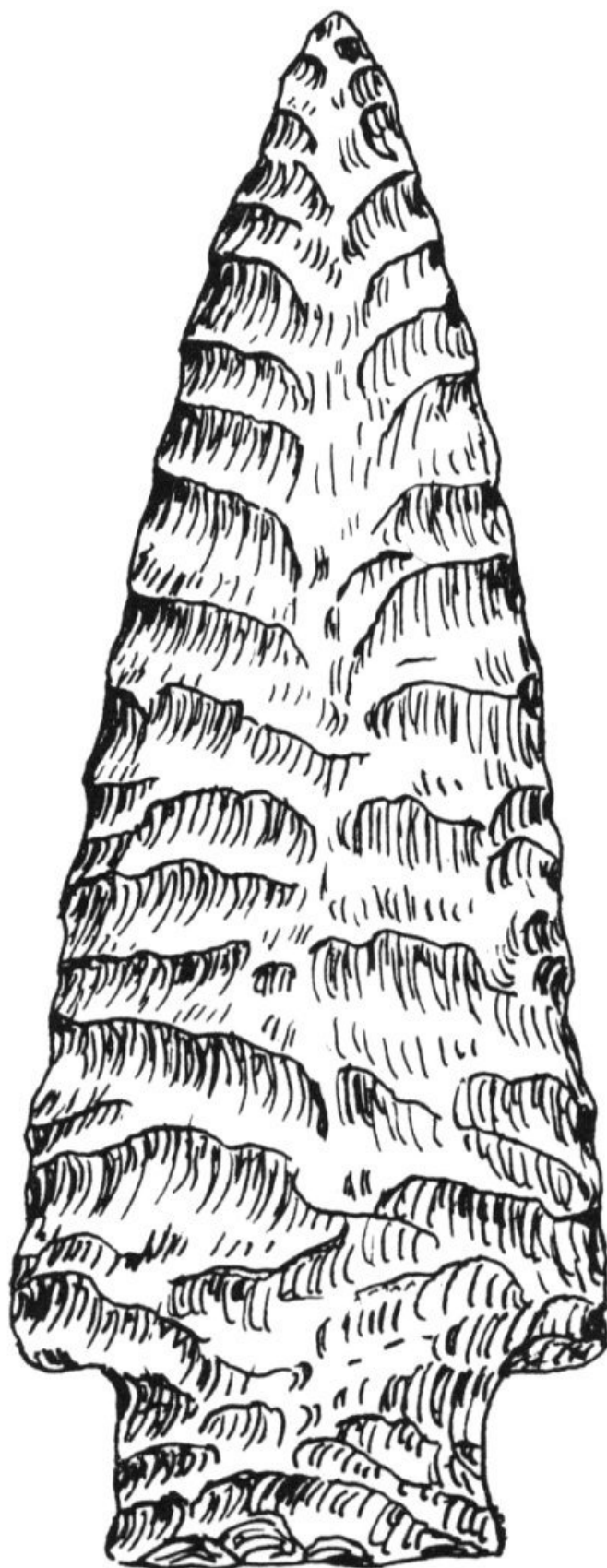
Meserve points are basically lanceolate-shaped. Upper lateral edges, which in the case of these points comprise almost three-quarters of the total point length, are typically straight. They are quite distinctive in being rather steeply alternately bevelled and they may be serrated. Bevelling is always on the right edge when viewed with the point tip forward; cross-section of the upper lateral edges at mid-section is rhomboidal. The unusually short lower lateral edges are straight or slightly concave. Meserve points are broadest either at the distal end of these edges, or at the base. The base is shallowly or markedly concave. Basal thinning is accomplished mainly by the removal of either one large flake or several narrower flakes parallel to the point long-axis, in a fashion reminiscent of the thinning of Plainview point bases. Less often, these scars are small and numerous. Thinning in some cases was by the removal of flakes at right angles to the point long-axis during initial retouch of the point faces. Average dimensions are: length 3.0 to 9.0 cm., width (base) 2.0 to 2.7 cm. Workmanship is generally good. Initial shaping as well as bevelling were likely by direct percussion. Initial flake scars are usually irregular. Basal thinning in some cases was probably by indirect percussion. Lateral edges may be secondarily pressure retouched. Smoothing is observed on both the lower non-bevelled lateral edges and in the concave base.

An estimated age of from 9,000 to 5,000 or even 4,000 years ago has been suggested for Meserve points. They are thus in part contemporaneous with, and have been found with, Plainview points. Meserve points are found throughout most of the Southwest, as far north as Canada, and in the Midwest states. They appear to be most common on the Llano Estacado.

SCOTTSBLUFF I POINTS



Scottsbluff I



Scottsbluff II



Eden

Projectile points of the type Scottsbluff were first defined on the basis of specimens recovered from a bison kill site near Scottsbluff, Nebraska.

The over-all shape of Scottsbluff I points is either lanceolate or triangular. Upper lateral edges are usually straight, but they may be faintly rounded. Characteristic of this type is a broad stem and faint shoulders. Parallel edges of the stem are straight; sometimes they may expand towards the base. The small shoulders, the widest part of these points, are cut inwards at right angles to the point long-axis. The stem base is straight or slightly rounded, very rarely slightly concave. Average dimensions range from about 5.0 to 15.0 cm. in length; width at the shoulders is from 2.4 to 3.5 cm., and width at the stem base 2.0 to 3.0 cm. Workmanship on most Scottsbluff I points is well executed. Initial shaping was carried out by direct or indirect percussion. Typical of many Scottsbluff I points are the longish parallel flake scars produced during this stage of retouch. They are at right angles to the point long-axis and are of the transverse type. On others patterning of flake scars is more irregular. Lateral edges of some may be secondarily pressure retouched. Both the stem lateral edges and base are smoothed.

Scottsbluff I points have a suggested age range of from 9,500 to about 7,000 years. They are known to occur not only in Nebraska, but in New Mexico, Texas, Oklahoma, Louisiana, Arkansas, Colorado, Wyoming, Montana, Washington, and in Saskatchewan, Alberta, and British Columbia.

SCOTTSBLUFF II POINTS

These points are probably of the same age and distribution as those of the Scottsbluff I type.

Scottsbluff II points are morphologically quite similar to those of the Scottsbluff I type with these exceptions: the lateral edges of the point body on Scottsbluff II points are straight, giving this portion of the point a definitely triangular outline; in addition, the point body is relatively broader at the shoulders, and the shoulders are more markedly offset from the stem than on Scottsbluff I points. Also, the former are generally shorter and thinner than those of the latter type.

EDEN POINTS

Eden points bear a great deal of resemblance to Scottsbluff points, with which they are sometimes found. Dissimilarities are pointed out below.

Eden points, though of the same basic shape, are much narrower in relation to over-all length than are Scottsbluff points. Most commonly, shoulders on Eden points are very faint indeed. Transverse facial flaking is much like that on Scottsbluff points. More frequently it is of the collateral type.

Probably the same age as the Scottsbluff type.

BROWNS VALLEY POINTS

Browns Valley points were initially discovered at a site near Browns Valley, Minnesota. They were associated with a human burial.



Browns Valley



Agate Basin

Browns Valley points are broad lanceolate in shape. Upper lateral edges are slightly rounded. Lower lateral edges are much less rounded, contracting very slightly to either a shallowly concave or straight base. These points are broadest at mid-section. Dimensions are: length 7.0 to 8.0 cm., width 3.0 cm. (full range indeterminate since few specimens known). 17

Workmanship is fine. Initial shaping may have been by direct or indirect percussion. The initial flake scars are parallel and run obliquely across the point faces; less often they are at right angles. Bases have been initially thinned by the removal of a series of small longish flakes mainly oblique to the point long-axis, to a lesser extent by the removal of vertical flakes. Basal edges are smoothed.

Browns Valley points are rare and have an estimated age of from 9,000 to 8,000 years. Similar points excavated in Wyoming have a comparable age of almost 8,000 years.

AGATE BASIN POINTS

Agate Basin points were first recognized at a bison kill site between Newcastle and Lusk in eastern Wyoming. The bison were probably of a recent type, but this is not certain.

Agate Basin points are slender lanceolate in shape. Both the upper and lower lateral edges are gently rounded. Lower lateral edges on some, however, may be nearly straight and contracting slightly to the base. They are usually broadest above mid-section. The base is either straight or rounded, although on some specimens it may be bluntly pointed. Length varies from 6.0 to 15.0 cm., width from 2.0 to about 3.0 cm. Workmanship is quite fine. Initial shaping was by direct or indirect percussion. Flake scars are consistently sub-parallel and at right angles to the point long-axis. Thinning of lateral edges was by pressure retouch. Lower lateral edges are smoothed, the base rarely so. Basal thinning was achieved during initial retouch and is at right angles to the point long-axis.

Although the age range of Agate Basin points is uncertain, it is probable that they belong to a later stage of the Early Man point sequence.

GOLONDRINA POINTS

Golondrina points were defined on the basis of specimens found at the Devils River Site on the Rio Grande in southwest Texas.

Golondrina points are of broad lanceolate shape. Upper lateral edges are usually markedly rounded. Lower lateral edges are characteristically 18 recurved, resulting in out-flaring basal corners of ears. The base is quite deeply concave. The broadest part is at mid-section or a little beyond, towards the tip. Average dimensions are: length 6.0 to 8.0 cm., maximum body width 2.3 to 3.2 cm., and base width 2.2 to 2.9 cm. Workmanship is fine. Initial and secondary shaping and thinning was by direct percussion. Initial flake scars are irregular. Pressure retouch of edges is not obvious. Essentially, vertical basal thinning was accomplished by the removal of either one or a series of irregular flakes. It should be noted that thinning scars on occasional specimens may be quite narrow and parallel to the point long-axis. Both the lower lateral edges and the basal concavity have been smoothed.



Golondrina

At the type site, Golondrina points have been dated to about 8,000 years ago. Plainview points may be associated with them. Golondrina points have been found in north central, central, and southwest Texas and northeast Mexico. It is likely they will be identified in other parts of the broader Southwest.

GYPSUM CAVE POINTS

Gypsum Cave points were discovered together with the remains of extinct ground sloth in a cave near Las Vegas, Nevada.

The over-all shape of Gypsum Cave points is elongate triangular. The straight upper lateral edges comprise virtually the entire length of specimens. The lower portion is characterized by a short stem formed by the abrupt sloping back of base edges from marked shoulders; the base is rounded. They are broadest from shoulder to shoulder. Average length of Gypsum Cave points is 5.0 to about 6.5 cm., and width about 3.0 cm. Flaking seems to have been produced by direct percussion. Initial flake scars are irregular. Secondary thinning of lateral edges is apparently absent. Stem edges were produced by multi-directional flaking. They are not smoothed.

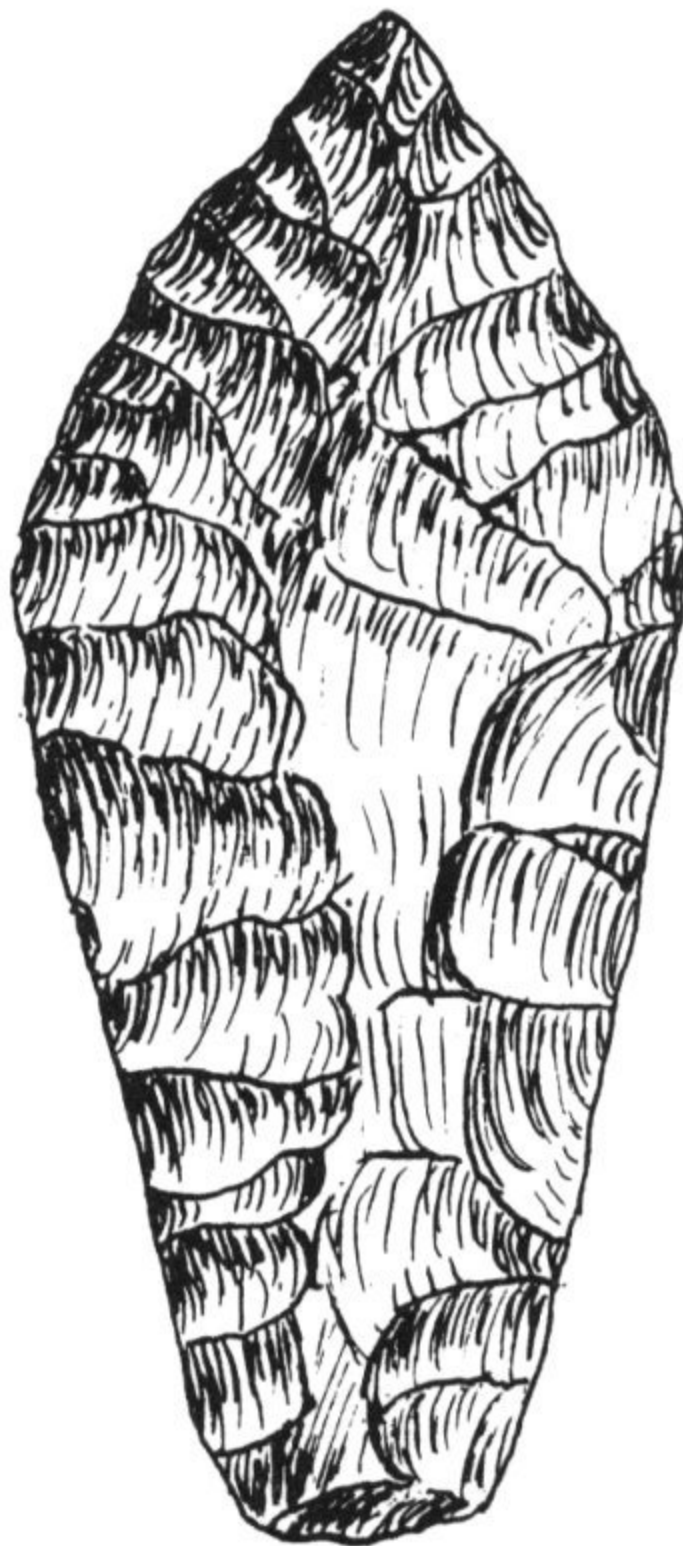


Gypsum Cave

The estimated age of Gypsum Cave points at the type site is from about 8,000 to 10,000 years. Points similar to these were found in a cave in the Manzano Mountains near Albuquerque and south of the Sandias. A specimen was recovered from the Lindenmeier Site in Colorado above the Folsom layer.

LAKE MOHAVE POINTS

Lake Mohave points were originally described from the Lake Mohave district of southeastern California. They were recovered from the surface and from ancient fossil beaches and terraces.



Lake Mohave

Lake Mohave points are slender lanceolate in shape with the broadest part above mid-section, about three-quarters up from the base. The upper lateral edges are slightly rounded. The over-all impression one has of these lower edges is that of a long tapering stem. Lake Mohave points may be faintly shouldered. Average dimensions are: length about 4.0 to 7.5 cm., width at the broadest part 2.5 to 3.0 cm. Workmanship, though well-achieved, is characterized by irregular initial flake scars, probably produced by direct percussion. Basal thinning was achieved by multi-directional flaking. 20

The estimated age range of Lake Mohave points is from about 7,000 to 9,000 years. It is probable this point type is related to similar types of the Southwest.

RIO GRANDE POINTS

Rio Grande points were first defined in 1942 on the basis of surface collections of lithic artifacts from the Upper Rio Grande Valley.



Rio Grande

Rio Grande points are basically lanceolate in shape. Upper lateral edges are straight to gently rounded. They are broadest slightly above mid-section. Lower lateral edges are set off from the upper portion of the point body by very small shoulders, so that this point must be considered stemmed. Stem edges are consistently straight and taper slightly to the base. The base is usually straight, less often it is slightly rounded or shallowly concave. Average dimensions are: length 6.0 cm., shoulder width 2.8 cm., base width 1.8 cm. Some larger specimens are known. Manufacture is by direct percussion. Initial flake scars on point faces are irregular. Basal thinning was accomplished in the main by multi-directional flaking. Stem edges may show secondary percussion or pressure retouch. Stem edges are well smoothed; the base may be slightly smoothed or not at all.

The age of Rio Grande points has not been definitely established nor have associations with other early point types been ascertained. They appear to be related to the Hell Gap and Lake Mohave types.

ESCOBAS POINTS

Escobas points, newly defined in this guide, were to all appearances first discovered and described at Pecos Pueblo, southeast of Santa Fe, New Mexico. Specimens excavated there evidently were collected by 21 inhabitants of the pueblos from nearby sites pre-dating the pueblo. This point type is named after Escobas Peak, near Pecos.



Escobas

Escobas points are lanceolate shaped and are consistently broadest immediately below mid-section. Upper lateral edges are usually straight, but they may be slightly rounded. Lower lateral edges are set off from the upper portion of the point body by very slight shoulders, so that this point must be considered as stemmed. Parallel to slightly expanding, stem edges are characteristically faintly concave between the shoulders and base edges. The base is always distinctly concave. Average dimensions: length 6.0 cm., shoulder width 2.0 cm., base width 1.7 cm. Workmanship is not refined. Initial flaking of point faces has been achieved by direct percussion; flake scars are irregular. Stem edges may show slight secondary retouch by either percussion or pressure. Basal thinning was accomplished primarily by the striking of irregular flakes from the basal concavity, to a lesser extent by removal of horizontal flakes from the base. Stem edges are well smoothed; the concave base is either slightly smoothed or not at all.

The relation of Escobas points to other Early Man points has not yet been determined. They do, however, resemble some other early types. Escobas points are broadly similar to Rio Grande points in shape, but they differ from them in the following respects: parallel to slightly expanding stem edges of Escobas points are consistently slightly concave between the shoulders and base, while stem edges on Rio Grande points are always straight and taper somewhat to the base. Escobas point stems are shorter, relative to overall point length than Rio Grande stems. Shoulders on Escobas are much fainter than on Rio Grande points; bases of Escobas points are, so far as known, always concave, whereas they are only occasionally so on Rio Grande points. Both, however, share the attribute of being well-smoothed on stem edges, and are either slightly smoothed or not at all on basal edges. Escobas points occur in the Pecos district, the Upper Rio Grande Valley, and, according to Stewart Peckham (personal communication), in the San Juan Basin of northwest New Mexico. They will likely be found to occur in other parts of the Southwest, in some areas in association with points of the Rio Grande type. 22

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The Denver Museum of Natural History: Figure 30, specimen 4; Figure 68, No. 5; Figure 70, Nos. 1, 2, and 3; Figure 71, Nos. 3 and 4; Figure 72, Nos. 1 and 3 from *Ancient Man in North America* by H. M. Wormington, Denver, 1957.

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American Antiquity: Figure 1b from *A New Point Type from Hell Gap Valley, Eastern Wyoming* by George A. Agogino, Vol. 26, No. 4, April 1961.

Museum of New Mexico Press
Santa Fe 1965

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Early Man
Projectile Points
in the Southwest

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